The Ramsey Seven-Cyclotomic Economy

A Constructive Framework for
Seven-District Economic Coordination and
Unification with an Eighteen-Participant Economy

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Abstract

This paper proposes a testable architecture for a seven-district economy whose algebraic reference structure is the seventh cyclotomic polynomial, and specifies an interface with a separate eighteen-participant economy. The identity $25 = 18+7$ records the combined count but does not make participants and districts interchangeable. The paper separates algebraic facts, modelling choices, hypotheses, and empirical questions. It presents a cyclic state-space model, a financial-planning-and-analysis layer, a welfare objective, a risk protocol, pseudocode, and an interdisciplinary research agenda. Connections to other fields are treated as modelling analogies requiring explicit mechanisms and evidence. No empirical validation is claimed.

The paper ends with "The End"

Keywords: cyclotomic polynomial; economic coordination; FP&A; econometrics; welfare economics; interoperability; resilience.

1 Purpose and epistemic discipline

The Ramsey Seven-Cyclotomic Economy (R7CE) is a proposed seven-district system for studying economic accounts, coordination, risk, and interaction with an eighteen-participant economy. Three categories must remain separate:

1. *Mathematical facts*, such as the degree and roots of a cyclotomic polynomial.
2. *Model assumptions*, such as district functions and feedback rules.
3. *Empirical propositions*, such as whether the system improves stability, welfare, or forecast accuracy.

An algebraic identity alone does not establish an economic causal mechanism. An analogy between factorization and coalition formation is a hypothesis-generating device, not evidence. The distinction between formal solvability, constructive procedures, and algorithmic implementation is useful methodological guidance [6].

2 Algebraic foundation

The m-th cyclotomic polynomial $\Phi_m(z)$ has as roots the primitive m-th roots of unity. For the prime number seven,

$$\Phi_7(z) = z^6 + z^5 + z^4 + z^3 + z^2 + z + 1. \tag{1}$$

Its roots are

$$\zeta_7^k, \qquad k = 1, \ldots, 6, \qquad \zeta_7 = \exp\left(\frac{2\pi i}{7}\right). \tag{2}$$

For $z \neq 1$, $z^7 - 1 = (z-1)\Phi_7(z)$. These are algebraic facts, not economic conclusions [1]. Let

$$\mathcal{D} = \{D_0, D_1, \ldots, D_6\}, \tag{3}$$

with indices interpreted modulo seven. A cyclic ordering is a modelling choice, not a consequence that economics must be cyclic.

3 District architecture

The functional assignments below are provisional categories for accounting and modelling, not claims that every economy naturally decomposes into these seven functions.

Table 1: Proposed district functions and observable indicators

District	Function	State	Candidate indicators
D_0	Production	$x_0(t)$	Output, productivity, capacity utilisation
D_1	Consumption	$x_1(t)$	Consumption, real disposable income
D_2	Investment	$x_2(t)$	Fixed investment, capital formation
D_3	Liquidity	$x_3(t)$	Credit, liquid assets, payment delays
D_4	Trade	$x_4(t)$	Imports, exports, terms of trade
D_5	Risk and resilience	$x_5(t)$	Stress losses, concentration, recovery time
D_6	Coordination	$x_6(t)$	Forecast revisions, compliance, decision latency

Each district may contain multiple sectors, agents, or institutions. A data dictionary should specify units, frequency, source, transformations, and missing-data treatment for every indicator.

4 Dynamics and propositions

Let $x_j(t)$ be the state of district D_j, $u_j(t)$ its control, and $\varepsilon_j(t)$ an exogenous disturbance. A general cyclic state-space model is

$$x_j(t+1) = f_j\big(x_j(t), x_{j-1 \pmod 7}(t), u_j(t), \varepsilon_j(t+1)\big). \tag{4}$$

Linearization around a candidate steady state gives

$$\Delta x(t+1) = A\Delta x(t) + B\Delta u(t) + \varepsilon(t+1). \tag{5}$$

The matrix A encodes own-district and cross-district effects. Cyclic sparsity is a candidate restriction to be tested.

Proposition 1 (conditional stability). For the deterministic linear system $\Delta x(t+1) = A\Delta x(t)$, the steady state is asymptotically stable if every eigenvalue of A has modulus strictly less than one.

Proof. The solution is $\Delta x(t) = A^t \Delta x(0)$. If the spectral radius $\rho(A) < 1$, then A^t converges to zero, so every initial deviation converges to zero. □ This standard result from linear-systems theory does not prove that any particular R7CE calibration is stable; its parameters must be estimated or bounded [2].

Proposition 2 (accounting consistency). If every internal transaction is recorded once as a source and once as a use with opposite signs, and all accounts use a common valuation basis, then internal net positions sum to zero, apart from external sectors, valuation changes, and statistical discrepancies.

Proof. Each internal transaction contributes $+v$ to one account and $-v$ to another. Summing over accounts cancels each such transaction. Any remaining balance comes from explicitly excluded items or discrepancies. □

5 Cyclotomic representation as a diagnostic

A seven-element vector can be represented by a discrete Fourier transform. For a scalar indicator x_j, define

$$\widehat{x}_k = \sum_{j=0}^{6} x_j \zeta_7^{-kj}, \qquad k = 0, \ldots, 6. \tag{6}$$

The inverse is

$$x_j = \frac{1}{7} \sum_{k=0}^{6} \widehat{x}_k \zeta_7^{kj}. \tag{7}$$

The zero-frequency component records the aggregate level; the remaining components describe cross-district variation across cyclic frequencies. For real-valued observations, $\widehat{x}_{7-k} = \overline{\widehat{x}_k}$. This is a diagnostic representation, not a causal explanation.

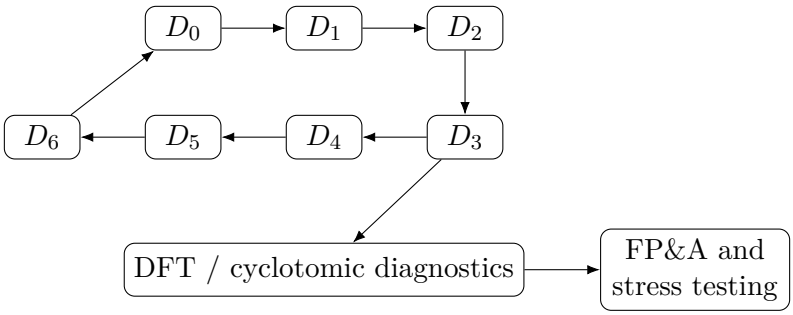


Figure 1: Proposed cyclic architecture and analysis pipeline. Arrows denote a candidate information structure, not established causal effects.

6 Coordination with the eighteen-participant economy

Let $\mathcal{E}_{18}$ be the separate eighteen-participant system with state $y(t)$, and let $\mathcal{E}_7$ denote R7CE with state $x(t)$. The combined count is

$$18 + 7 = 25. \tag{8}$$

This arithmetic does not make participants and districts commensurable. Their relationship must be defined by variables and interface contracts rather than by concatenating labels. A

general coupled model is

$$x(t+1) = F_7(x(t), u_7(t), M_{18\to 7}y(t), \eta_7(t+1)), \tag{9}$$
$$y(t+1) = F_{18}(y(t), u_{18}(t), M_{7\to 18}x(t), \eta_{18}(t+1)). \tag{10}$$

The interface maps must specify dimensions, units, timing, permissions, and aggregation. Interoperability can permit cooperation without requiring systems to become identical [5].

Table 2: Minimum interface contract

Element	Required specification
Variables	Names, units, sign conventions, valuation basis
Timing	Frequency, time zone, release lag, revision policy
Aggregation	Mapping participant records to district measures
Governance	Authority to propose, approve, execute, and reverse decisions
Privacy	Access rights, retention limits, permitted uses
Failure handling	Missing feeds, stale data, conflicts, fallback mode
Auditability	Versioned inputs, transformation logs, reproducible outputs

7 Financial planning, welfare, and risk

The financial-planning-and-analysis (FP&A) layer converts district data into forecasts, budgets, scenarios, and variance analysis. For district j, let $R_j(t)$, $C_j(t)$, and $I_j(t)$ denote revenue, expenditure, and investment:

$$N_j(t) = R_j(t) - C_j(t) - I_j(t). \tag{11}$$

Internal transfers must be reconciled so they are not counted as new consolidated income. Forecast errors should feed into later assumptions.

A welfare-oriented planner may solve

$$\max_{\{u_7(t),u_{18}(t)\}} \quad \mathbb{E}\left[\sum_{t=0}^{T} \beta^t W(x(t), y(t))\right] \tag{12}$$
$$\text{subject to} \quad \text{coupled dynamics, resource constraints, and institutional rules,} \tag{13}$$
$$u_7(t) \in \mathcal{U}_7, \qquad u_{18}(t) \in \mathcal{U}_{18}, \tag{14}$$

where $0 < \beta \leq 1$ and W is an explicitly selected social-welfare criterion. Distributional weights, rights, environmental effects, and intergenerational concerns are normative choices and should be disclosed [3].

For a loss variable L, Value-at-Risk is a quantile and Expected Shortfall summarizes tail loss beyond that quantile. Stress tests should include correlated shocks, liquidity shortages, supply-chain interruption, model misspecification, and interface failure. No single statistic completely captures systemic safety [4].

8 Econometric and statistical evaluation

Compare at least four candidates: (i) a cyclic seven-district model; (ii) a fully connected seven-node model; (iii) a model without cross-district feedback; and (iv) the pre-existing planning or forecasting baseline. Use rolling or expanding-window out-of-sample evaluation. Metrics may

include mean absolute error, root mean squared error, calibration, forecast revisions, constraint violations, recovery time, and defensible welfare proxies. Causal claims require a credible counterfactual and attention to confounding, reverse causality, structural breaks, and measurement error.

For forecast error $e_t = y_t - \widehat{y}_t$,

$$\mathrm{MAE} = \frac{1}{T}\sum_{t=1}^{T} |e_t|, \tag{15}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T} e_t^2}. \tag{16}$$

Report uncertainty intervals, robustness checks, and failed tests. Better prediction alone does not establish that a policy caused better welfare outcomes [8].

9 Interdisciplinary interpretation

Each field must contribute a defined mechanism rather than a decorative analogy.

Table 3: Cross-disciplinary contributions and limits

Field	Potential contribution	Boundary condition
Mathematics and logic	Invariants, feasibility, proofs	Algebraic symmetry is not economic symmetry
Physics and control	Feedback, stability, spectral diagnostics	Agents may adapt to the model
Chemistry and biology	Network resilience, resource flows	Analogies require explicit mechanisms
Psychology and behavioural science	Expectations, bounded rationality, coordination costs	Do not infer mental states from aggregates alone
Psychiatry and public health	Ethical assessment of stress and human impacts	Do not diagnose populations from macro indicators
Linguistics and journalism	Stable terms, provenance, uncertainty reporting	Headlines must not outrun evidence
Political science and IR	Institutions, bargaining, commitments	Formal coordination does not imply consensus
Diplomacy and geopolitics	Verification, negotiation, escalation safeguards	Models do not guarantee peace
Warfare economics and security	Resource diversion and disruption scenarios	Emphasize resilience and civilian protection
Computer science and AI/ML	Forecasting, anomaly detection, reproducibility	Validate, monitor, and retain human accountability
Cybersecurity	Least privilege, integrity checks, recovery	Stable equations can still be compromised digitally
Religion and philosophy	Reflection on justice, duties, restraint, welfare	Traditions are diverse; no single reading is assumed

The inclusion of a discipline does not establish that its theories are validated in R7CE. Journalism contributes a useful epistemic rule: distinguish observation, inference, forecast, and advocacy. Ethics requires individual and distributional impacts to remain visible in aggregate planning.

10 Algorithm and pseudocode

The initial implementation should prioritize transparent data handling over complexity.

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INPUT:
district observations X; participant observations Y
data dictionary; interface contract; baseline models

1. Validate units, timestamps, permissions, and missingness.
2. Reconcile accounts; flag unexplained discrepancies.
3. Estimate baseline and cyclic seven-district models.
4. Compute DFT diagnostics and stability measures.
5. Generate FP&A forecasts and adverse scenarios.
6. Exchange only authorized, validated interface variables.
7. Evaluate forecasts and stress outcomes out of sample.
8. Record uncertainty, failures, versions, and rationale.
9. Require human approval for consequential policy changes.
10. Preserve audit trails; revise only after documented review.

OUTPUT:
forecasts, reconciliations, risk indicators, comparisons,
limitations, and reproducible decision records
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Production code should include unit tests, schema validation, version control, dependency management, access logging, backup-restoration tests, and independent review. AI/ML systems should not autonomously change consequential policy settings without defined authority, monitoring, and rollback.

11 Cybersecurity and governance

The interface is a risk surface. Threats include data poisoning, unauthorized access, replayed or stale records, compromised credentials, denial of service, and silent code changes. Controls should include least privilege, strong authentication, integrity checks, protected data transfer and storage, tested backups, incident response, and separation between analysis and execution. Governance must specify data ownership, approval authority, conflict resolution, emergency procedures, and post-incident review. Mathematical stability cannot compensate for compromised data or institutions.

12 Limitations and research agenda

This paper is an architecture proposal, not an empirical validation. The district assignments are design choices; cyclic topology is a testable restriction; the DFT is diagnostic rather than causal; and $25 = 18 + 7$ does not prove institutional compatibility. Next steps are to:

1. define district indicators and accounting conventions;
2. specify and test the interface contract;

3. prove model-specific feasibility and stability under stated assumptions;
4. implement a reproducible simulation with transparent parameter ranges;
5. compare against simpler and non-cyclic baselines;
6. evaluate distributional effects, stress resilience, and governance failures;
7. publish code, synthetic test data, and unsuccessful as well as successful tests.

13 Conclusion

R7CE is a constructive research programme: seven functional districts, an algebraic reference structure based on Φ_7, a candidate cyclic state-space model, an FP&A and welfare layer, and an explicit interoperability contract with a distinct eighteen-participant system. The identity $25 = 18 + 7$ motivates joint design without erasing the systems' different units. Mathematical claims can be proved; economic claims must be operationalized and tested; normative choices must be disclosed. This separation is the foundation for a credible implementation.

References

[1] L. C. Washington, *Introduction to Cyclotomic Fields*, 2nd ed., Springer, 1997.

[2] K. Ogata, *Modern Control Engineering*, 5th ed., Prentice Hall, 2010.

[3] A. Sen, *Collective Choice and Social Welfare*, expanded ed., Harvard University Press, 2017.

[4] Basel Committee on Banking Supervision, *Principles for the Management of Credit Risk*, Bank for International Settlements, 2000.

[5] International Organization for Standardization, *ISO/IEC/IEEE 42010: Systems and Software Engineering—Architecture Description*, ISO, 2011.

[6] S. Ghosh, *The Complete Treatise on the General Sextic: With a Rigorous Existence Proof for Ghosh's Sextic Parametrisation and Interdisciplinary Perspectives*, manuscript, 2026. Its claims are a methodological reference and require independent verification before being treated as established theorems.

[7] F. P. Ramsey, "On a Problem of Formal Logic," *Proceedings of the London Mathematical Society*, s2-30(1), 264–286, 1930.

[8] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed., Wiley, 2015.

Glossary

Cyclotomic polynomial
: Polynomial whose roots are primitive roots of unity of a specified order.

District state
: Measurements describing a district's economic condition at a given time.

DFT
: Discrete Fourier transform, representing a finite sequence in frequency components.

FP&A
: Financial planning and analysis: budgeting, forecasting, scenarios, and variance review.

Interface map
: A specified transformation between variables in distinct systems.

Interoperability
: Capacity of distinct systems to exchange and use information under defined conventions.

Model risk
: Risk that assumptions, data, implementation, or use are inadequate.

Out-of-sample evaluation
: Assessment using observations not used to fit a model.

Resilience
: Capacity to absorb disturbance, maintain essential functions, and recover under specified conditions.

Welfare function
: Explicit criterion for comparing social outcomes, with normative assumptions disclosed.

The End